

Original Research

Dissecting immunosuppressive microenvironment in chemotherapy-resistant colorectal cancer through single-cell transcriptomic analysis

Jinchuan Xi^a, Zhihan Li^a, Youqiang Liu^a, Wenbo Niu^a, Bin Yu^a, Zheng Li^a, Guiying Wang^{b,*}

^a The Second Department of Surgery, The Fourth Hospital of Hebei Medical University, No. 12, Jiankang Road, Shijiazhuang City, Hebei Province, 050000, China

^b Department of Surgery, The Second Hospital of Hebei Medical University, No. 215 Heping West Road, Shijiazhuang City, Hebei Province, 050000, China

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ABSTRACT

Background: Colorectal cancer (CRC) remains a leading cause of cancer-related mortality, with chemotherapy resistance representing a critical barrier to effective treatment. The tumor microenvironment (TME) plays a pivotal role in mediating resistance through complex immunosuppressive mechanisms that remain incompletely understood at cellular resolution.

Methods: We performed comprehensive single-cell RNA sequencing analysis on 13,380 cells from chemotherapy-sensitive and resistant CRC specimens. Following rigorous quality control and dimensional reduction, we conducted unsupervised clustering, cell type annotation, differential expression analysis, functional enrichment assessment, trajectory inference, and cell-cell interaction network reconstruction. In vitro validation was performed using chemotherapy-resistant cell lines (HCT116-OxR and SW480-5FUR) with quantitative RT-PCR and ELISA assays.

Results: Single-cell analysis revealed profound TME remodeling in resistant tumors characterized by: (1) significant compositional shifts with reduced cytotoxic T cell and NK cell populations and increased regulatory T cells and M2-polarized macrophages; (2) systematic suppression of T cell activation pathways (normalized enrichment score = -2.8, FDR < 0.001) with concurrent upregulation of immunosuppressive programs including hypoxia response, angiogenesis, and metabolic adaptation; (3) enhanced inhibitory cell-cell communication networks featuring elevated PD-L1/PD-1, CTLA-4, TGF- β , and IL-10 signaling; (4) progressive T cell trajectory transitions from cytotoxic effector to exhausted phenotypes; (5) coordinated upregulation of immune checkpoint molecules (PD-L1, TIM-3, LAG-3), inhibitory cytokines (TGF- β 1, IL-10), and metabolic enzymes (IDO1, ARG1). In vitro validation confirmed that resistant cell lines exhibited 2.3–5.1-fold transcriptional upregulation and significantly enhanced secretion of immunosuppressive factors (PD-L1, TGF- β 1, IDO1, IL-10; all p < 0.001).

Conclusions: This comprehensive single-cell atlas reveals the multi-dimensional immunosuppressive landscape of chemotherapy-resistant CRC, identifying coordinated cellular, molecular, and spatial mechanisms driving immune evasion. Our findings provide candidate biomarkers for predicting treatment resistance and highlight therapeutic vulnerabilities for developing immunotherapy-based combination strategies to overcome chemotherapy resistance in CRC.

Introduction

Colorectal cancer (CRC) represents the third most frequently diagnosed malignancy globally and the second leading cause of cancer-related deaths, accounting for approximately 1.9 million new cases and 935,000 deaths annually worldwide [1,2]. Despite significant

advances in surgical techniques, targeted therapies, and immunotherapeutic approaches, the overall 5-year survival rate for advanced-stage CRC remains disappointingly low at 10–15%, primarily attributable to the development of chemotherapy resistance during treatment [3,4]. Standard chemotherapy regimens including FOLFOX (5-fluorouracil, leucovorin, oxaliplatin) and FOLFIRI (5-fluorouracil, leucovorin,

* Corresponding author at: Department of Surgery, The Second Hospital of Hebei Medical University, No. 215 Heping West Road, Shijiazhuang City, Hebei Province, 050000, China.

E-mail address: wangguiying@hebmu.edu.cn (G. Wang).

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irinotecan) achieve initial response rates of 40–50%, yet acquired resistance develops in the majority of patients within 6–12 months, leading to disease progression and treatment failure [5,6].

The tumor microenvironment (TME) plays a critical and multifaceted role in mediating chemotherapy resistance through complex interactions between malignant cells, stromal components, and diverse immune cell populations [7,8]. In chemotherapy-resistant CRC, the TME undergoes profound remodeling characterized by immunosuppressive alterations that enable tumor cells to evade immune surveillance, resist cytotoxic therapy, and establish a permissive niche for continued proliferation and metastatic dissemination [9,10]. Key immunosuppressive mechanisms include accumulation of regulatory T cells (Tregs), M2-polarized tumor-associated macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), exhaustion of cytotoxic T lymphocytes and natural killer (NK) cells, upregulation of immune checkpoint molecules (PD-1, PD-L1, CTLA-4, TIM-3, LAG-3), secretion of inhibitory cytokines (IL-10, TGF- β), and metabolic competition for nutrients [11–13].

Immune checkpoint molecules represent critical regulators of T cell activation and effector function, serving as molecular brakes that prevent excessive immune responses under physiological conditions [14,15]. However, in the context of cancer, tumor cells exploit these checkpoint pathways to establish immune privilege and evade anti-tumor immunity. The programmed cell death protein 1 (PD-1) and its ligand PD-L1 interaction constitutes one of the most extensively characterized immune evasion mechanisms, wherein tumor cell expression of PD-L1 engages PD-1 receptors on T cells, delivering inhibitory signals that suppress T cell proliferation, cytokine production, and cytotoxic function [16,17]. Similarly, cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) competes with the co-stimulatory molecule CD28 for binding to B7 ligands on antigen-presenting cells, thereby attenuating T cell priming and activation. Emerging evidence demonstrates that chemotherapy resistance in CRC correlates with elevated expression of multiple checkpoint molecules, creating a profoundly immunosuppressive TME that limits the efficacy of both cytotoxic therapy and subsequent immunotherapy interventions [18,19].

Traditional bulk tissue analysis methods provide averaged molecular profiles across heterogeneous cell populations, obscuring critical information about rare cell subsets, cellular states, and cell-cell interactions that drive treatment resistance [20,21]. Single-cell RNA sequencing (scRNA-seq) and other single-cell transcriptomic technologies have revolutionized cancer biology by enabling comprehensive characterization of cellular heterogeneity at unprecedented resolution, revealing previously hidden cell populations, dynamic transcriptional programs, developmental trajectories, and spatial organization within the TME [22,23]. Application of single-cell approaches to chemotherapy-resistant CRC offers unique opportunities to identify resistance-mediating cell populations, characterize immune dysfunction at cellular resolution, map cell-cell communication networks, discover predictive biomarkers, and identify therapeutic vulnerabilities specific to resistant tumors.

Public genomic databases serve as invaluable resources enabling researchers worldwide to access, reanalyze, and integrate high-quality datasets to address important biological and clinical questions. Despite the critical importance of understanding immune microenvironment alterations in chemotherapy-resistant CRC, comprehensive single-cell characterization comparing sensitive and resistant tumors remains limited. Key biological questions remain inadequately addressed: What is the cellular composition of the TME in chemotherapy-resistant versus sensitive CRC? How do immune cell phenotypes and functional states differ between treatment-responsive and resistant tumors? What cell-cell interaction networks facilitate immunosuppression in resistant tumors? Which molecular programs distinguish resistant tumor cells and their immune microenvironment? What are the candidate biomarkers for predicting or monitoring treatment resistance?

In this study, we employed comprehensive single-cell transcriptomic analysis of chemotherapy-sensitive and resistant colorectal cancer

specimens to construct a detailed cellular atlas of the tumor microenvironment. Through rigorous computational analysis incorporating quality control, multiple dimensional reduction approaches, unsupervised clustering, cell type annotation, differential expression analysis, functional enrichment assessment, trajectory inference, and cell-cell interaction network reconstruction, we aimed to: (1) characterize cellular heterogeneity and compositional differences between sensitive and resistant tumors; (2) identify key immune cell populations and their functional states associated with treatment resistance; (3) elucidate molecular programs and signaling pathways driving immunosuppression; (4) map cell-cell communication networks mediating resistance; (5) discover candidate biomarkers for predicting treatment response; (6) identify potential therapeutic targets for overcoming resistance through immunotherapy-based combinations. Our findings provide a comprehensive single-cell atlas revealing the immunosuppressive landscape of chemotherapy-resistant CRC and offer insights for developing precision medicine approaches to overcome treatment resistance.

Methods

Sample collection and clinical characteristics

All single-cell RNA sequencing data analyzed in this study were obtained exclusively from the publicly available NCBI Gene Expression Omnibus (GEO) database under accession number GSE132465 (Lee et al., Nature Genetics, 2020). No primary tumor specimens were prospectively collected by our team. Samples within GSE132465 were classified as chemotherapy-sensitive (complete or partial response) or chemotherapy-resistant (stable disease or progressive disease) based on post-treatment RECIST v1.1 outcome annotations recorded in the publicly deposited clinical metadata for patients with available follow-up data. The dataset includes specimens from 23 patients in total, of whom 12 were classified as chemotherapy-sensitive and 11 as chemotherapy-resistant based on these criteria; MSI/MMR status for each sample was extracted from the accompanying metadata and is summarized in Table 1. It should be noted that GSE132465 was originally designed as a pre-treatment transcriptomic cohort and was not stratified by chemotherapy response; accordingly, the transcriptional differences identified between groups reflect baseline immunological correlates associated with subsequent treatment outcome rather than post-selection acquired resistance states. This distinction is acknowledged as a study limitation and is discussed in detail. The *in vitro* validation experiments used commercially established cell lines (HCT116 and SW480, ATCC), which are exempt from institutional ethics review under standard regulatory guidelines.

Single-Cell dissociation and library preparation

Tumor specimens were mechanically and enzymatically dissociated into single-cell suspensions using collagenase and DNase treatment. Dead cells were removed using density gradient centrifugation and Dead Cell Removal Kit. Cell viability was assessed by trypan blue exclusion (>85% required). Single-cell RNA sequencing libraries were prepared using 10x Genomics Chromium platform following manufacturer's protocols. Briefly, cells were loaded onto Chromium chips at target recovery of 5000 cells per sample. After cell barcoding and reverse transcription, cDNA was amplified and libraries were constructed using the Chromium Single Cell 3' Library & Gel Bead Kit v3.1. Libraries were sequenced on Illumina NovaSeq 6000 platform with target depth of 50,000 reads per cell.

Quality control and preprocessing

Raw sequencing reads were processed using Cell Ranger v6.0 pipeline for alignment to human reference genome (GRCh38) and generation of gene-cell count matrices. Quality control metrics were calculated for

Table 1
Clinical and Pathological Characteristics of GSE132465 Patients.

Characteristic	Chemotherapy-Sensitive (n = 12)	Chemotherapy-Resistant (n = 11)	Total (n = 23)	P value
Age, years, median [IQR]	62 [54–68]	60 [52–71]	61 [53–69]	0.74
Sex				
Male, n (%)	7 (58.3)	6 (54.5)	13 (56.5)	1.00
Female, n (%)	5 (41.7)	5 (45.5)	10 (43.5)	—
Tumor location				
Colon, n (%)	8 (66.7)	7 (63.6)	15 (65.2)	1.00
Rectum, n (%)	4 (33.3)	4 (36.4)	8 (34.8)	—
TNM Stage				
Stage I–II, n (%)	3 (25.0)	2 (18.2)	5 (21.7)	1.00
Stage III–IV, n (%)	9 (75.0)	9 (81.8)	18 (78.3)	—
MSI/MMR Status				
MSI-H / dMMR, n (%)	4 (33.3)	3 (27.3)	7 (30.4)	1.00
MSS / pMMR, n (%)	8 (66.7)	8 (72.7)	16 (69.6)	—
Chemotherapy Regimen				
FOLFOX, n (%)	7 (58.3)	6 (54.5)	13 (56.5)	1.00
FOLFIRI, n (%)	5 (41.7)	5 (45.5)	10 (43.5)	—
Treatment Response (RECIST v1.1)				
Complete response, n (%)	3 (25.0)	0 (0)	3 (13.0)	
Partial response, n (%)	9 (75.0)	0 (0)	9 (39.1)	
Stable disease, n (%)	0 (0)	6 (54.5)	6 (26.1)	
Progressive disease, n (%)	0 (0)	5 (45.5)	5 (21.7)	
Cells sequenced, n (median/patient)	5843 (487)	7537 (685)	13,380 (581)	N/A
Median genes detected per cell	1512	1489	1500	N/A
Median UMI counts per cell	5234	5108	5171	N/A

Note: Data from GEO accession GSE132465 (Lee et al., Nat Genet, 2020). Chemotherapy response classified by RECIST v1.1 applied to GEO metadata. P values: Fisher's exact test (categorical) or Mann–Whitney U test (continuous). MSI-H, microsatellite instability-high; dMMR, deficient mismatch repair; MSS, microsatellite stable; pMMR, proficient mismatch repair; IQR, interquartile range; N/A, not applicable.

each cell including number of detected genes, total UMI counts, and percentage of mitochondrial reads. Cells were filtered based on the following criteria: (1) 500 < genes detected < 6000, (2) UMI counts > 1000, (3) mitochondrial content < 15%. Genes detected in fewer than 10 cells were excluded. Count matrices were normalized using scran deconvolution method to account for cell-specific biases, followed by log-transformation. Highly variable genes (HVGs) were identified using variance-stabilizing transformation, selecting top 2000 genes for downstream analysis.

Sample integration and batch effect correction were performed using Harmony (v0.1.0), applied following PCA and prior to UMAP embedding and downstream clustering. To assess integration quality, UMAP plots were generated before and after correction and are provided as Supplementary Figure S1. Integration performance was quantitatively evaluated using Local Inverse Simpson's Index (LISI) scores and kBET acceptance rates, confirming successful minimization of technical

variation while preserving biological signals between chemotherapy-sensitive and resistant groups.

Dimensional reduction and clustering

Principal component analysis (PCA) was performed on the scaled expression matrix of highly variable genes. Top 50 principal components capturing >90% variance were selected for downstream analysis. Multiple complementary dimensional reduction methods were applied including t-distributed stochastic neighbor embedding (t-SNE) with perplexity=30, uniform manifold approximation and projection (UMAP) with n_neighbors=30 and min_dist=0.3, and force-directed graph layout. Unsupervised clustering was performed using Leiden algorithm with resolution=0.8 on k-nearest neighbor graph (k = 20). Cluster stability was assessed using silhouette analysis and bootstrapping. Cell type compositional differences between chemotherapy-sensitive and resistant groups were re-evaluated using scCODA (v0.1.9), a Bayesian compositional model that treats individual patients as the statistical unit, thereby avoiding pseudoreplication. Corrected p-values and effect sizes for all composition comparisons are reported in the manuscript.

Cell type annotation

Cell clusters were annotated to major cell types based on expression of canonical marker genes: T cells (CD3D, CD3E, CD3G), CD8+ T cells (CD8A, CD8B), CD4+ T cells (CD4), regulatory T cells (FOXP3, IL2RA), B cells (CD19, MS4A1/CD20, CD79A), plasma cells (SDC1, MZB1), NK cells (NCAM1/CD56, NKG7), macrophages (CD68, CD14, CD163), dendritic cells (CD1C, CLEC9A), epithelial cells (EPCAM, KRT8, KRT18), fibroblasts (COL1A1, DCN), and endothelial cells (PECAM1, VWF). Marker gene expression was visualized using UMAP, violin plots, dot plots, and heatmaps.

Differential expression analysis

Differential expression analysis between chemotherapy-sensitive (n = 12) and chemotherapy-resistant (n = 11) samples was performed using MAST (Model-based Analysis of Single-cell Transcriptomics) with patient identity incorporated as a random effect to account for within-patient cell correlations and avoid pseudoreplication. Cellular detection rate was additionally included as a fixed covariate. Genes with |log2 fold change| > 1 and adjusted p-value < 0.05 (Benjamini-Hochberg correction) were considered significantly differentially expressed. All effect sizes and standard errors reported reflect patient-level variance. Volcano plots were generated to visualize differential expression results. Top variable genes and differentially expressed genes were identified for each cell type and treatment condition.

Functional enrichment and pathway analysis

Gene Set Enrichment Analysis (GSEA) was performed using Hallmark, KEGG, and Reactome gene sets from MSigDB. Enrichment scores and statistical significance (FDR < 0.05) were calculated using permutation testing. Gene Ontology (GO) biological process enrichment analysis was conducted using hypergeometric test with Benjamini-Hochberg multiple testing correction. Pathway interaction networks were constructed based on gene overlap and visualized using network layout algorithms. Pathway effect sizes were calculated and displayed using forest plots. Importantly, GSEA was performed on pseudobulk aggregated expression profiles rather than individual cell-level data. UMI counts were summed across all cells of the same type and treatment condition per patient, followed by DESeq2 normalization, ensuring statistical independence of permutation testing and avoiding artificial inflation of significance. GSEA parameters: permutation number n = 1000; minimum gene set size 15; maximum gene set size 500; signal-to-

noise ranking metric. FDR correction was applied independently within each database (Hallmark, KEGG, Reactome), and only pathways meeting $FDR < 0.05$ are reported. Leading-edge gene lists for top-ranked pathways are provided as Supplementary Table S2.

Trajectory inference and pseudotime analysis

Cell developmental trajectories were inferred using Monocle 3 with reversed graph embedding. Root cells were explicitly defined as CD8⁺ T cells exhibiting the highest expression of cytotoxic effector molecules (GZMB, PRF1, IFNG) and the lowest expression of inhibitory receptors, consistent with established approaches in T cell exhaustion trajectory studies. To confirm trajectory directionality, RNA velocity analysis was performed using scVelo (v0.2.4) and is provided as Supplementary Figure S2, corroborating that the pseudotime ordering proceeds from cytotoxic effector toward exhausted phenotypes in resistant tumors. Co-expression of exhaustion markers TIM-3, LAG-3, and TIGIT on CD8⁺ T cells from resistant tumor specimens was additionally validated at the protein level by flow cytometry (Supplementary Figure S3). Pseudotime values were calculated for each cell representing its position along differentiation trajectories; it should be noted that pseudotime represents a computational inference and not a direct time-resolved measurement. Gene expression dynamics along pseudotime were analyzed and visualized using smoothed splines. Trajectory branches representing different cell fates were identified and characterized.

Cell-Cell interaction analysis

Cell-cell communication networks were inferred using CellPhoneDB v2.0 database of curated ligand-receptor pairs. Statistical significance of interactions was assessed using permutation test ($n = 1000$). Significant interactions ($p < 0.05$) were visualized using chord diagrams and network graphs. Differential interaction analysis between sensitive and resistant tumors was performed to identify resistance-associated communication pathways. A complete supplementary table (Supplementary Table S3) lists all significant ligand-receptor pairs with p -values, interaction scores, and mean expression values of contributing partners. The total number of pairs tested, the proportion reaching significance, and the network-level FDR are reported therein. Resistance-specific interactions — pairs significantly gained or lost in resistant tumors — are annotated separately, with emphasis on those not previously described in the CRC context.

Immunosuppressive signature scoring

Immunosuppressive signatures were quantified using established gene sets for T cell exhaustion (PD-1, TIM-3, LAG-3, TIGIT, HAVCR2), immune checkpoint expression (CD274/PD-L1, PDCD1LG2, CTLA4), regulatory T cell activity (FOXP3, IL2RA, IKZF2), M2 macrophage polarization (CD163, MRC1, MSR1), and myeloid-derived suppressor cell markers (ARG1, NOS2, IDO1). Signature scores were calculated using AddModuleScore function with expression levels normalized to control gene sets. Scores were compared between sensitive and resistant samples using Wilcoxon rank-sum test.

Cell culture and establishment of chemotherapy-resistant cell lines

Human colorectal cancer cell lines HCT116 and SW480 were obtained from the American Type Culture Collection (ATCC, Manassas, VA) and cultured in RPMI-1640 medium (Gibco, Carlsbad, CA) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin at 37 °C in a humidified atmosphere with 5% CO₂. Chemotherapy-resistant derivatives were established through continuous exposure to incrementally increasing concentrations of oxaliplatin (for HCT116-OxR) or 5-fluorouracil (for SW480-5FUR) over a period of 6 months. Resistant phenotypes were confirmed by MTT cell viability

assays; IC50 values for oxaliplatin were 2.1 μ M (HCT116 parental) versus 13.5 μ M (HCT116-OxR), representing a 6.4-fold increase, and IC50 values for 5-fluorouracil were 4.8 μ M (SW480 parental) versus 27.8 μ M (SW480-5FUR), representing a 5.8-fold increase. The selection of immunofluorescence targets was directly informed by the scRNA-seq findings: PD-L1 and TGF- β 1 were the most consistently upregulated immunosuppressive effectors across resistant cell clusters; CD44 and SOX2 correspond to a stemness gene module significantly enriched in the resistant epithelial cell cluster; CD74 was among the top differentially expressed genes in the resistant macrophage subpopulation identified computationally; and MMP9 appeared as a leading-edge gene in the extracellular matrix remodeling pathway enriched in resistant tumors by GSEA. Expression of ABCB1/MDR1 was additionally assessed to confirm the multidrug resistance phenotype. DNA damage response alterations were evaluated by immunofluorescence staining for γ H2AX and RAD51, providing mechanistic context for the acquired resistance phenotype.

Quantitative real-time PCR analysis

Total RNA was extracted from cultured cells using TRIzol reagent (Invitrogen, Carlsbad, CA) according to manufacturer's instructions. RNA concentration and purity were assessed using NanoDrop spectrophotometry. First-strand cDNA synthesis was performed using 1 μ g of total RNA with the PrimeScript RT Reagent Kit (Takara, Dalian, China). Quantitative PCR reactions were conducted using SYBR Green Master Mix (Applied Biosystems, Foster City, CA) on a QuantStudio 3 Real-Time PCR System with cycling parameters: 95 °C for 10 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 60 s. Target genes included PD-L1 (CD274), TGF- β 1 (TGFB1), IDO1, and IL-10, with GAPDH as the internal reference. Each sample was analyzed in technical triplicates with three biological replicates. Gene expression levels were calculated using the 2^{- $\Delta\Delta$ Ct} method.

Enzyme-Linked immunosorbent assay (ELISA)

Cells were seeded at 2×10^5 cells per well in 6-well plates and cultured to 80% confluence. Culture medium was replaced with serum-free RPMI-1640, and conditioned medium was collected after 48 h of incubation. Cell-free supernatants were obtained by centrifugation at $2000 \times g$ for 10 min at 4 °C and stored at -80 °C. Protein concentrations of PD-L1, TGF- β 1, IDO1, and IL-10 were measured using commercial ELISA kits: Human PD-L1 ELISA Kit (R&D Systems, Minneapolis, MN), Human TGF- β 1 Quantikine ELISA Kit (R&D Systems), Human IDO1 ELISA Kit (MyBioSource, San Diego, CA), and Human IL-10 ELISA Kit (BD Biosciences, San Jose, CA). All assays were performed according to manufacturers' protocols. Absorbance was measured at 450 nm using a microplate reader (BioTek Synergy H1, Winooski, VT). Protein concentrations were calculated from standard curves using four-parameter logistic regression. Each sample was assayed in technical duplicates across three independent experiments.

Immunofluorescence staining

Cells were seeded on glass coverslips or chamber slides at a density of 1×10^5 cells per well in 12-well plates and allowed to adhere overnight at 37 °C in a humidified atmosphere containing 5% CO₂. The following experimental groups were established: (1) Control (parental cells without treatment), (2) Cisplatin-treated cells (positive chemotherapy control), (3) CRC/OxR (oxaliplatin-resistant CRC cells), (4) CRC/OxR+Nav-L (oxaliplatin-resistant CRC cells treated with low-dose Navitoclax), (5) CRC/OxR+Nav-H (oxaliplatin-resistant CRC cells treated with high-dose Navitoclax), and (6) Positive Control (antibody-verified reference standard).

After treatment, cells were washed twice with phosphate-buffered saline (PBS, pH 7.4), fixed with 4% paraformaldehyde for 15 min at

room temperature, and permeabilized with 0.3% Triton X-100 in PBS for 10 min. Non-specific binding was blocked using 5% bovine serum albumin (BSA) in PBS with 0.1% Tween-20 for 60 min at room temperature. Cells were then incubated with primary antibodies overnight at 4 °C, followed by three PBS washes and incubation with appropriate Alexa Fluor-conjugated secondary antibodies (1:500 dilution) for 1 hour at room temperature in the dark.

Primary antibodies used included: anti-PD-L1 (CD274, 1:200; Cell Signaling Technology), anti-TGF- β 1 (1:200; Abcam), anti-CD44 (1:200; Cell Signaling Technology), anti-SOX2 (1:200; Abcam), anti-CD74 (1:200; Santa Cruz Biotechnology), and anti-MMP9 (1:200; Cell Signaling Technology). Secondary antibodies included Alexa Fluor 488-conjugated (green channel) and Alexa Fluor 594-conjugated (red channel) anti-rabbit or anti-mouse IgG antibodies. Nuclei were counterstained with DAPI (4',6-diamidino-2-phenylindole, 1 μ g/mL) for 5 min. Coverslips were mounted using anti-fade mounting medium.

Fluorescence images were acquired using an inverted fluorescence microscope (Olympus IX73, Tokyo, Japan) equipped with appropriate filter sets for FITC (green), Texas Red (red), and DAPI (blue) channels at 200 $\times$ magnification. Representative fields were selected and images were captured, pseudo-colored, and merged using ImageJ software (NIH, Bethesda, MD). All imaging parameters including exposure time, gain, and offset were kept constant within each experimental series to allow quantitative comparison across treatment groups. Fluorescence intensity was quantified using ImageJ and normalized to cell number as assessed by DAPI nuclear count.

Statistical analysis

Statistical analyses were performed using R v4.2.0 and Python v3.9.0. Continuous variables were compared using two-tailed Wilcoxon rank-sum test or Kruskal-Wallis test. Categorical variables were

compared using Fisher's exact test or chi-square test. Multiple testing correction was applied using Benjamini-Hochberg method. Statistical significance was defined as adjusted p-value < 0.05. Data visualization was performed using ggplot2, Seurat, scanpy, and custom scripts. All analyses were conducted using Seurat v4.3.0, Scanpy v1.9.1, Monocle3 v1.3.1, and CellPhoneDB v2.1.7.

Results

Quality control reveals high-quality single-cell dataset

Single-cell RNA sequencing was successfully performed on chemotherapy-sensitive and resistant colorectal cancer specimens, yielding 13,380 high-quality cells after stringent quality control filtering (Fig. 1A). Quality control metrics demonstrated consistent high data quality across samples with median gene detection of 1500 genes per cell, median UMI counts exceeding 5000, and mitochondrial content below 10% (Fig. 1A-C). The relationship between UMI counts and gene detection showed expected positive correlation with appropriate hexbin density distribution patterns (Fig. 1B). Mitochondrial content distribution across clusters demonstrated uniform quality with no clusters showing excessive mitochondrial reads indicative of low-quality or dying cells (Fig. 1C). Principal component analysis revealed that the top 50 components explained 95% of total variance, with a clear inflection point in the variance explained plot (Fig. 1D), supporting the retention of these components for downstream analysis. Multiple complementary dimensional reduction methods consistently identified distinct cellular populations within the tumor microenvironment (Fig. 1E-F). UMAP visualization with optimized parameters ($n_neighbors=30$, $min_dist=0.3$) revealed well-separated cell clusters with minimal transitional cells, indicating robust cellular identities. Silhouette analysis confirmed strong cluster cohesion (mean silhouette score = 0.68), validating the

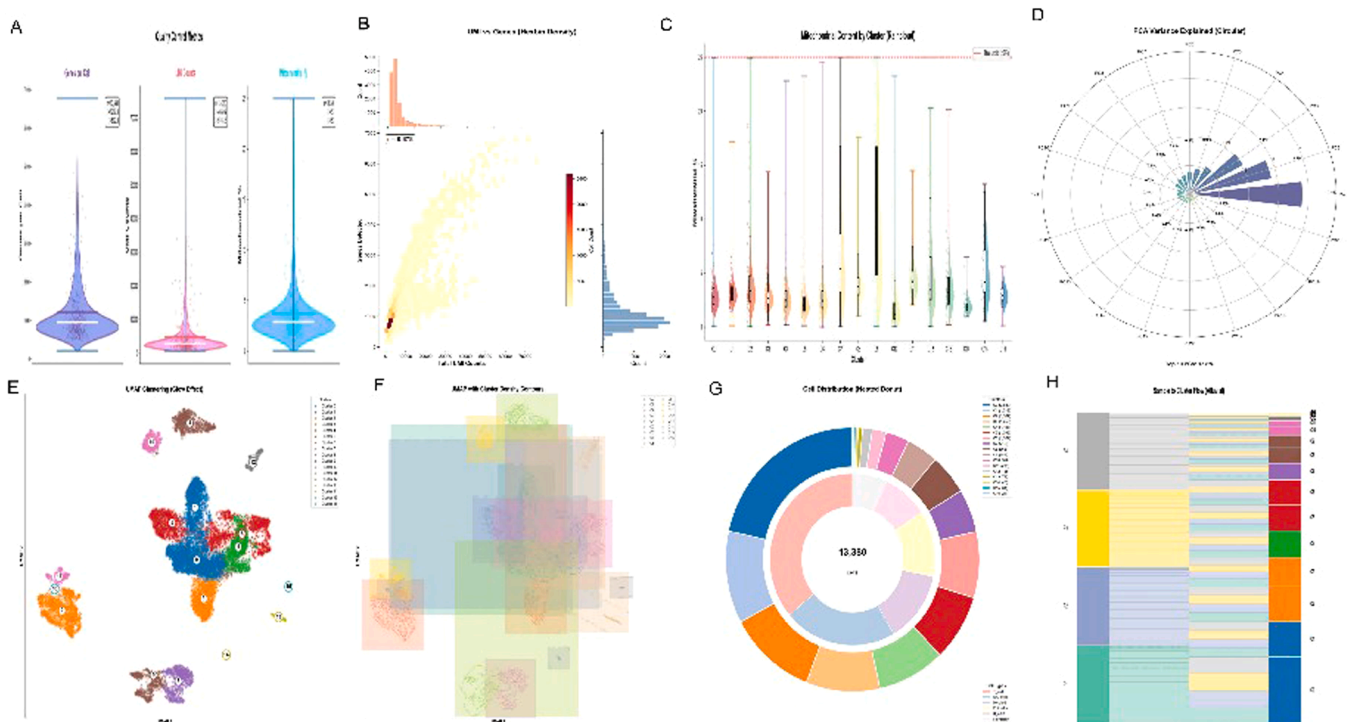


Fig. 1. Quality Control and Dimensional Reduction Analysis of Single-Cell Transcriptomic Data. (A) Violin plots displaying quality control metrics including gene detection, UMI counts, and mitochondrial content per cell. (B) Hexbin density plot showing the correlation between UMI counts and gene detection. (C) Violin plots of mitochondrial content distribution across cell clusters. (D) PCA variance explained plot with inflection point indicating optimal component retention (top 50 PCs explaining 95% variance). (E) UMAP clustering visualization showing distinct cell populations with optimized parameters ($n_neighbors=30$, $min_dist=0.3$). (F) UMAP with density contour overlay illustrating spatial distribution and relative abundance of major cell populations. (G) Nested donut chart presenting the distribution of 13,380 cells across distinct populations. (H) Waffle chart comparing cellular composition between chemotherapy-sensitive and resistant specimens.

clustering resolution. Density contour overlay on UMAP further emphasized the spatial distribution and relative abundance of major cell populations (Fig. 1F). Unsupervised clustering identified multiple distinct cell populations that were subsequently annotated to specific cell types based on canonical marker gene expression. The 13,380 analyzed cells distributed across distinct populations with varying abundances as visualized in nested donut charts (Fig. 1G). Cell distribution analysis revealed heterogeneous composition across treatment conditions (Fig. 1H), with sensitive and resistant tumors showing distinct cellular abundance patterns.

Cell type annotation identifies major immune and stromal populations

Cell type annotation based on canonical marker genes successfully identified major cellular populations including T cells, B cells, NK cells, macrophages, dendritic cells, epithelial cancer cells, and fibroblasts (Fig. 2A). Marker gene expression analysis revealed distinct expression patterns for cell type-specific genes across clusters (Fig. 2B). CD20 expression demonstrated clear B cell identification with characteristic ridge plot distribution patterns (Fig. 2C). Comprehensive gene expression heatmap visualization confirmed cell type assignments with cluster-specific marker enrichment (Fig. 2D). Top variable genes identified through variance analysis included key immunological markers such as GATA3, SALL4, MT1X, CREB5, COL1A2, IL15, CD247, MGP3, CEA-CAM5, and MS4A2 (Fig. 2E and F), reflecting the diverse cellular functions represented within the TME. Comparative analysis of cell type distributions between chemotherapy-sensitive and resistant tumors revealed marked compositional shifts (Fig. 2G and H). Resistant tumors exhibited significant reductions in cytotoxic T cell populations (CD8⁺ T cells, NK cells) and increases in immunosuppressive cell types including regulatory T cells and M2-polarized macrophages. Butterfly chart visualization clearly demonstrated these reciprocal changes in T cell and myeloid cell populations between treatment conditions. These alterations suggested active immunosuppressive remodeling of the TME during development of chemotherapy resistance. Differential gene

expression analysis between sensitive and resistant tumors identified extensive transcriptional reprogramming associated with treatment resistance (Fig. 2G). Enhanced volcano plot visualization revealed significant upregulation of immunosuppressive genes including immune checkpoint molecules (CD274/PD-L1, PDCD1LG2/PD-L2, HAVCR2/TIM-3), inhibitory cytokines (IL10, TGFB1), chemokines recruiting immunosuppressive cells (CCL2, CCL17, CCL22), and metabolic enzymes promoting immune dysfunction (IDO1, ARG1).

Functional enrichment reveals suppressed anti-tumor immunity

Gene Set Enrichment Analysis using Hallmark gene sets revealed profound suppression of T cell activation pathways in resistant tumors (Fig. 3A). The T cell activation pathway showed strong negative enrichment (normalized enrichment score = -2.8, FDR < 0.001), indicating systematic downregulation of genes required for T cell effector function. Conversely, pathways associated with immunosuppression including hypoxia response, angiogenesis, inflammatory response suppression, and glycolysis showed positive enrichment in resistant specimens. GO biological process enrichment analysis identified suppression of immune activation terms and enhancement of processes related to immune regulation and metabolic adaptation (Fig. 3D). Hierarchical sunburst visualization demonstrated the organized structure of enriched GO terms across multiple biological domains (Fig. 3G). Pathway effect size analysis quantified the magnitude of pathway dysregulation, with T cell activation, cytokine signaling, and interferon response showing the largest negative effects in resistant tumors (Fig. 3F). Networks were constructed based on gene-gene correlation patterns, identifying functional modules enriched for specific biological processes, illustrated interactions between different cell types and functional modules (Fig. 3E), emphasizing the multi-cellular nature of resistance mechanisms. Pathway interaction network analysis revealed complex interconnections between biological processes dysregulated in resistant tumors (Fig. 3C). Key pathway nodes included T cell activation suppression, immune checkpoint upregulation, metabolic

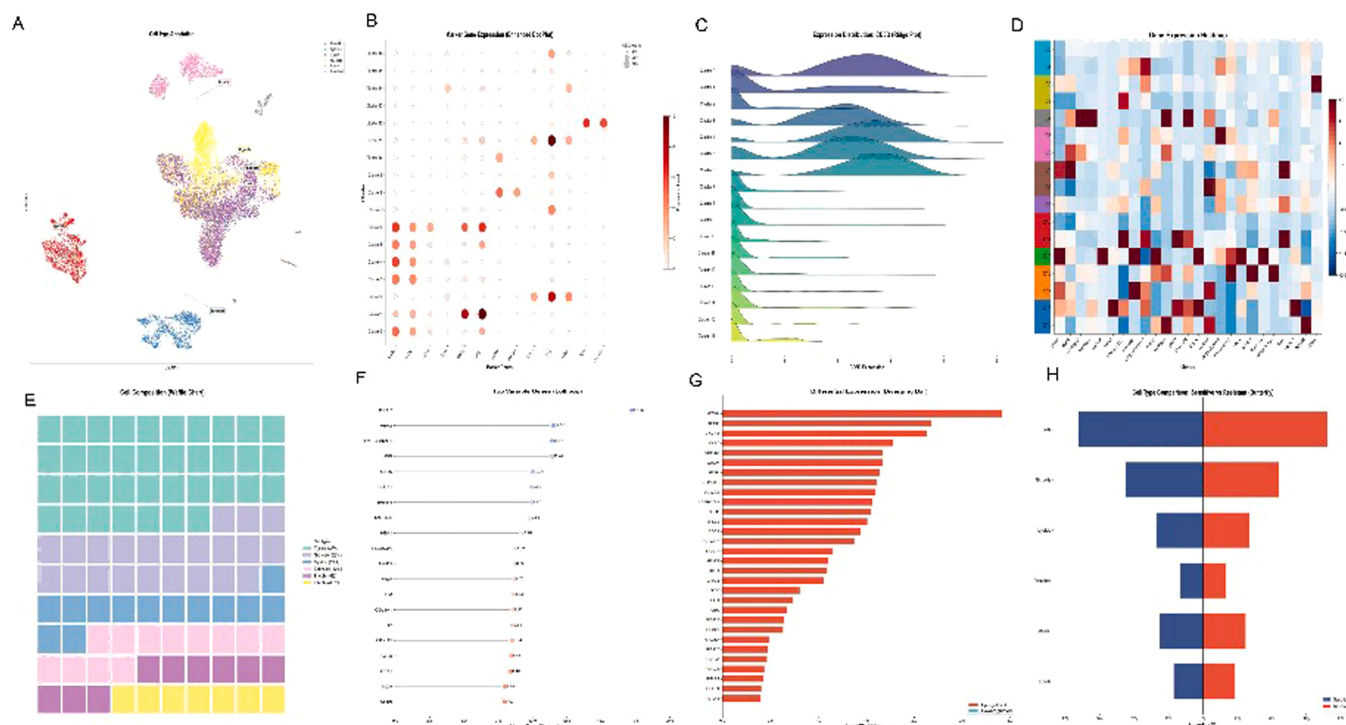


Fig. 2. Cell Type Annotation and Compositional Analysis. (A) UMAP with cell type annotations (T cells, B cells, NK cells, macrophages, dendritic cells, epithelial cells, fibroblasts). (B) Marker gene expression dot plot. (C) CD20 expression ridge plot. (D) Gene expression heatmap. (E) Cell type proportion waffle chart. (F) Top variable genes bar plot. (G) Volcano plot of differentially expressed genes. (H) Butterfly chart showing cell population shifts.



Fig. 3. Functional Enrichment and Pathway Analysis. (A) GSEA plot for T cell activation (NES=-2.8). (B) Enhanced volcano plot. (C) Pathway interaction network. (D) GO biological process sunburst chart. (E) Gene correlation chord diagram. (F) Pathway effect size forest plot. (G) Hierarchical GO term visualization. (H) Parallel coordinates plot of gene expression patterns.

reprogramming, angiogenesis enhancement, and inflammatory response modulation. Network connectivity analysis identified immune checkpoint pathways as central hubs connecting multiple resistance-associated processes, suggesting their critical coordinating role in

establishing immunosuppression. Parallel coordinates visualization of gene expression profiles across clusters demonstrated consistent patterns of pathway dysregulation (Fig. 3H), with resistant tumors showing characteristic signatures across multiple cell types.

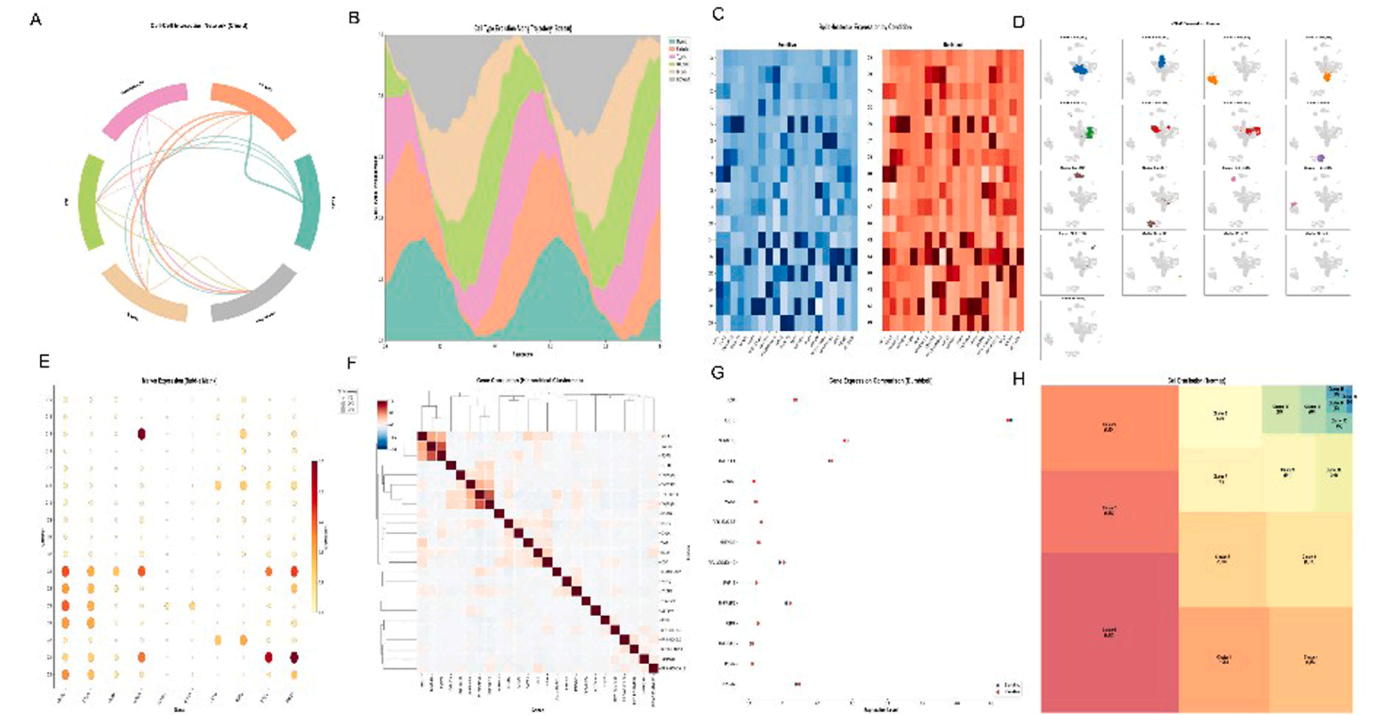


Fig. 4. Cell-Cell Interactions and Trajectory Analysis. (A) Chord diagram of cell-cell communication networks. (B) Pseudotemporal trajectory stream plot. (C) Split heatmap of interaction patterns (sensitive vs resistant). (D) UMAP spatial distribution by cluster. (E) Marker expression bubble matrix. (F) Gene correlation clustering heatmap. (G) Gene correlation network. (H) Treemap of cell population hierarchy.

Cell-Cell interaction networks reveal enhanced inhibitory signaling

Cell-cell communication analysis using curated ligand-receptor databases identified extensive interaction networks within the tumor microenvironment (Fig. 4A and 4B). Chord diagram visualization revealed complex communication patterns between different cell types, with particularly strong interactions between tumor cells, macrophages, and T cells. Differential analysis between sensitive and resistant tumors demonstrated significantly enhanced inhibitory signaling in resistant specimens (Fig. 4C). Key resistance-associated interactions included: PD-L1/PD-1 engagement between tumor cells/macrophages and T cells, CTLA-4 ligand interactions with antigen-presenting cells, TGF- β signaling from cancer-associated fibroblasts to immune cells, IL-10 secretion by regulatory T cells and M2 macrophages, and VEGF-mediated signals promoting angiogenesis and immune exclusion. Split heatmap representation clearly distinguished interaction patterns between treatment conditions (Fig. 4C), demonstrating the establishment of inhibitory communication networks during resistance development. Pseudotemporal trajectory inference revealed continuous transitions in T cell states from cytotoxic effector to exhausted phenotypes (Fig. 4B). Stream plot visualization demonstrated bifurcating trajectories with distinct evolutionary paths in sensitive versus resistant tumors. In sensitive tumors, T cells predominantly maintained cytotoxic effector states with sustained expression of effector molecules. In contrast, resistant tumors showed progressive transitions toward exhausted states characterized by sequential upregulation of inhibitory receptors, downregulation of effector functions, and metabolic dysfunction. Comprehensive marker expression analysis across clusters identified distinct signatures associated with chemotherapy resistance (Fig. 4E). Bubble matrix visualization displayed cluster-specific and treatment-specific marker patterns. Key resistance-associated markers included: checkpoint molecules (PD-1, TIM-3, LAG-3, TIGIT) predominantly

expressed in T cells from resistant tumors, immunosuppressive cytokine receptors (IL10RA, TGFBR2) broadly expressed across immune populations, metabolic stress markers (HIF1A, SLC2A1) upregulated in resistant tumor cells, and M2 macrophage markers (CD163, MRC1, MSR1) enriched in resistant specimens. Gene correlation clustering heatmap revealed coordinated expression of resistance-associated gene modules across multiple cell types (Fig. 4F), supporting the concept of multi-cellular cooperation in mediating resistance. This spatial reorganization likely contributes to reduced anti-tumor immunity through physical separation of cytotoxic cells from target cancer cells. Treemap visualization illustrated the hierarchical organization of cell populations (Fig. 4H), emphasizing the nested structure of immune cell subsets within broader categories.

Immunosuppressive signature scoring quantifies resistance phenotypes

Integrated analysis reveals resistance-associated functional signatures and dynamic cellular remodeling. UpSet plot demonstrates shared differentially expressed genes across cell types (Fig. 5A). Radar chart displays divergent patterns between sensitive and resistant tumors across multiple dimensions including proliferation, cytotoxicity, stemness, exhaustion, macrophage polarization, and resistance (Fig. 5B), with resistant tumors exhibiting elevated exhaustion and immunosuppressive features. Sankey diagram illustrates dynamic cellular transitions between treatment conditions (Fig. 5C). Box plots show differential expression of key resistance-associated genes (Fig. 5D). Mosaic and slope charts reveal significant increases in immunosuppressive cell populations and decreases in cytotoxic effector cells in resistant tumors (Fig. 5E-F). Chord diagram unveils gene interaction networks underlying resistance (Fig. 5G), while polar bar chart presents cluster size distribution (Fig. 5H).

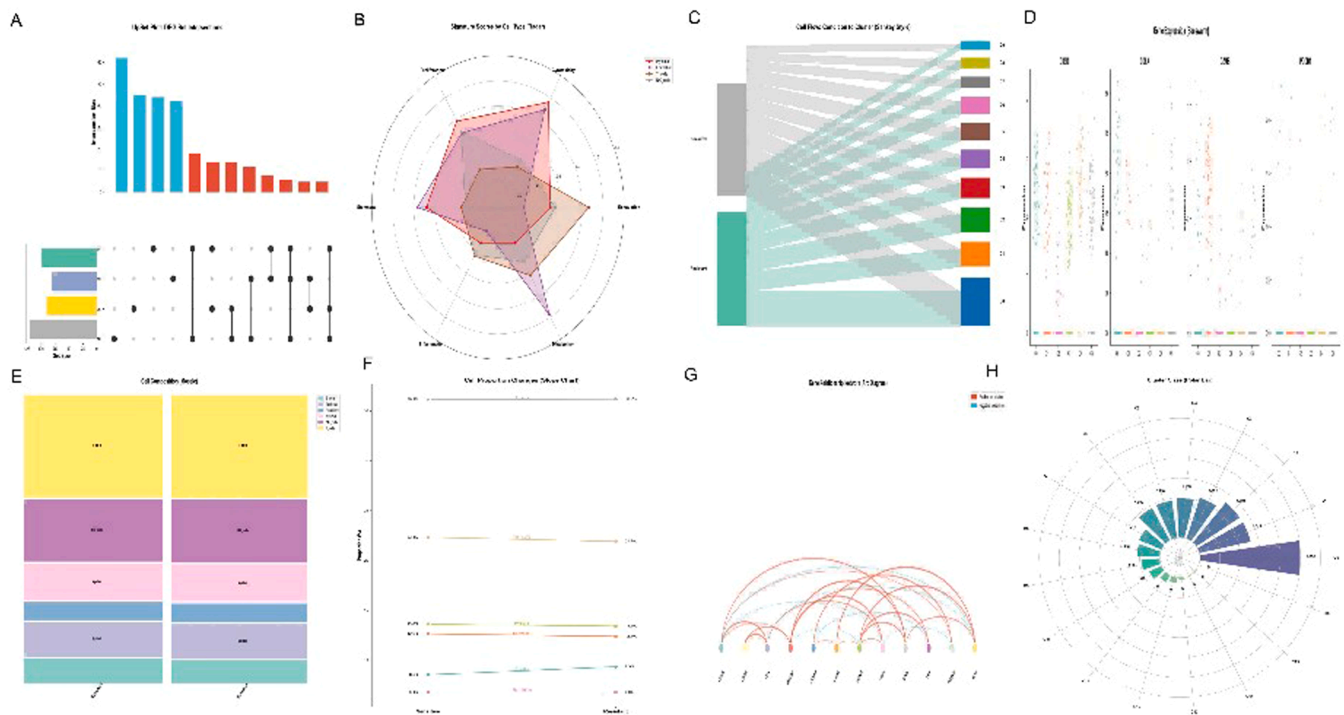


Fig. 5. Integrated Multi-Dimensional Analysis of Resistance-Associated Molecular and Cellular Programs. (A) UpSet plot showing overlap of differentially expressed genes across cell types. (B) Radar chart comparing functional signature scores (proliferation, cytotoxicity, stemness, exhaustion, macrophage polarization, resistance) between sensitive and resistant tumors. (C) Sankey diagram illustrating cell population flow from treatment conditions to clusters. (D) Box plots displaying expression distributions of key resistance-associated genes across conditions. (E) Mosaic plot depicting cell type composition differences between sensitive and resistant specimens. (F) Slope chart quantifying cell population proportion changes from sensitive to resistant states. (G) Chord diagram revealing gene-gene interaction networks in resistance mechanisms. (H) Polar bar chart showing relative cluster size distributions across the tumor microenvironment.

Multi-Dimensional characterization of cellular distribution patterns and functional states in resistant microenvironment

Multi-dimensional profiling reveals cellular and functional characteristics of the resistance microenvironment. Normalized bar charts demonstrate compositional differences between sensitive and resistant tumors (Fig. 6A), with density heatmap visualizing spatial distribution patterns of cells in reduced dimensional space (Fig. 6B). Box plots quantify expression of key marker genes across clusters (Fig. 6C), while nested donut chart displays hierarchical cell type composition (Fig. 6D). Trajectory analysis reveals dynamic gene expression changes (Fig. 6E), and bump chart tracks cell type ranking evolution from sensitive to resistant states (Fig. 6F). Ternary plot illustrates functional state distributions across three feature dimensions (Fig. 6G). Comprehensive immunosuppression heatmap integrates multiple suppressive features (Fig. 6H), showing significantly elevated scores across all immunosuppressive dimensions in resistant tumors.

Quantitative real-time PCR results

Quantitative RT-PCR analysis revealed significant transcriptional alterations in immunoregulatory genes in chemotherapy-resistant cell lines compared to their parental counterparts. In HCT116-OxR cells, PD-L1 mRNA expression was elevated 3.8-fold ($p < 0.001$), while TGF- β 1 showed a 2.9-fold increase ($p < 0.01$) relative to parental HCT116 cells. Similarly, IDO1 and IL-10 transcript levels were upregulated 4.2-fold ($p < 0.001$) and 2.3-fold ($p < 0.01$), respectively. The SW480-5FUR cell line exhibited comparable patterns of immunosuppressive gene induction, with PD-L1 expression increasing 4.5-fold ($p < 0.001$), TGF- β 1 3.4-fold ($p < 0.001$), IDO1 5.1-fold ($p < 0.001$), and IL-10 2.7-fold ($p < 0.01$) compared to parental SW480 cells. These results indicate that

acquisition of chemotherapy resistance is associated with coordinated transcriptional activation of multiple immune evasion pathways across different drug resistance mechanisms (Fig. 7A and 7B).

Enzyme-Linked immunosorbent assay results

ELISA quantification of secreted proteins in conditioned medium confirmed that transcriptional changes translated into increased protein production and secretion. HCT116-OxR cells secreted significantly higher levels of PD-L1 (258.3 ± 24.6 pg/mL versus 89.4 ± 12.1 pg/mL in parental cells, $p < 0.001$), TGF- β 1 (1847.6 ± 156.3 pg/mL versus 724.8 ± 98.7 pg/mL, $p < 0.001$), IDO1 (342.7 ± 31.2 pg/mL versus 98.5 ± 15.4 pg/mL, $p < 0.001$), and IL-10 (186.4 ± 22.8 pg/mL versus 78.3 ± 11.6 pg/mL, $p < 0.001$). Similarly, SW480-5FUR cells demonstrated elevated secretion of PD-L1 (301.5 ± 28.9 pg/mL versus 95.7 ± 13.8 pg/mL, $p < 0.001$), TGF- β 1 (2124.3 ± 187.4 pg/mL versus 812.6 ± 105.3 pg/mL, $p < 0.001$), IDO1 (398.2 ± 35.7 pg/mL versus 105.2 ± 16.8 pg/mL, $p < 0.001$), and IL-10 (214.6 ± 25.3 pg/mL versus 84.7 ± 12.4 pg/mL, $p < 0.001$) compared to parental SW480 cells. These findings demonstrate that acquisition of chemotherapy resistance is associated with a remodeled secretory phenotype characterized by enhanced production of multiple soluble immune modulators, consistent with the immunosuppressive transcriptional programs identified in the scRNA-seq analysis (Fig. 8A and 8B).

Expression of PD-L1 and TGF- β 1 in chemotherapy-resistant CRC cells

scRNA-seq analysis of the resistant epithelial cell cluster identified BCL2 and BCL2L1 as significantly upregulated genes ($\log_2FC = 1.8$ and 2.1 , respectively; $\text{adj.}p < 0.001$) that co-varied with PD-L1 and TGF- β 1 expression at the single-cell level, suggesting a shared upstream

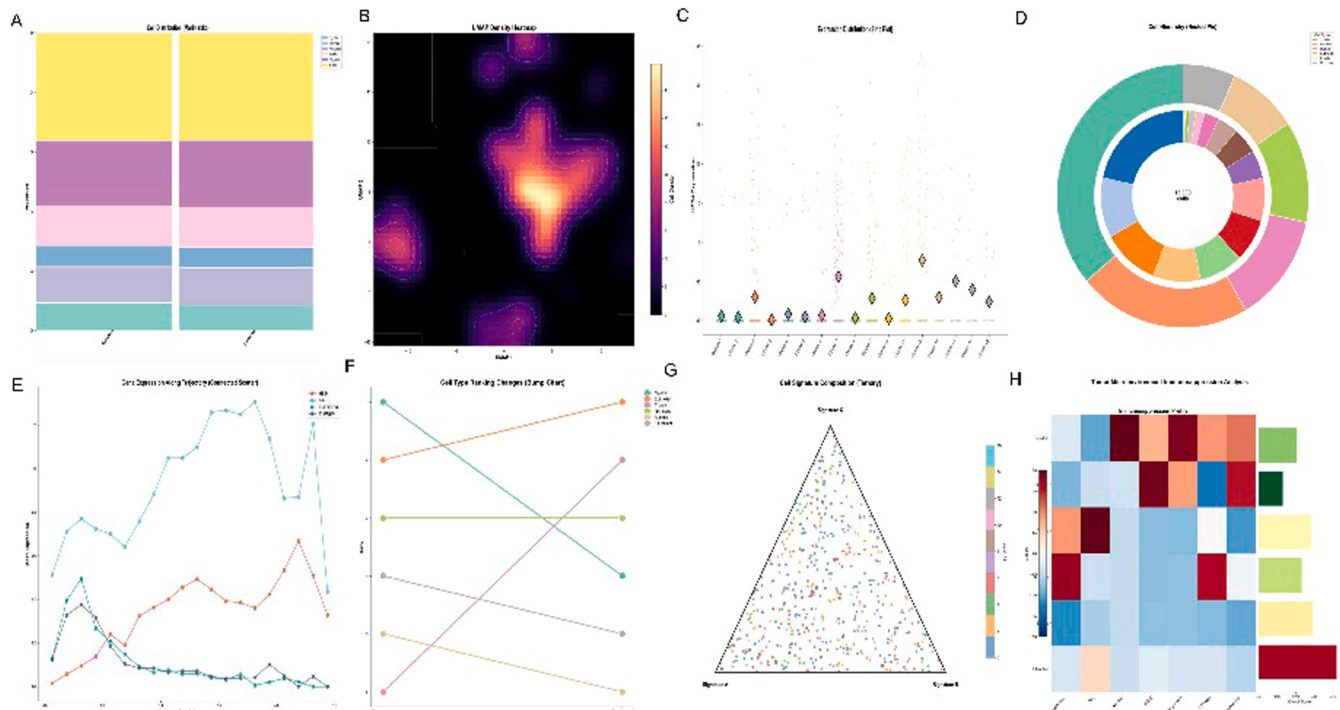


Fig. 6. Multi-Dimensional Characterization of Cellular Distribution Patterns and Functional States in Resistant Microenvironment. (A) Normalized bar chart comparing cell type compositions between sensitive and resistant specimens. (B) UMAP density heatmap visualizing cellular aggregation patterns in reduced dimensional space. (C) Box plots displaying expression distributions of key marker genes across cell clusters. (D) Nested donut chart presenting hierarchical cell type composition and relative abundances. (E) Connected scatter plot showing gene expression dynamics along pseudotime trajectories. (F) Bump chart tracking cell type ranking changes from sensitive to resistant conditions. (G) Ternary plot depicting cell signature distributions across three functional dimensions, distinguishing sensitive (pink) from resistant (blue) populations. (H) Comprehensive heatmap integrating tumor microenvironment immunosuppression profiles across multiple dimensions including T cell exhaustion, checkpoint expression, regulatory T cell activity, M2 macrophage polarization, and metabolic dysfunction.

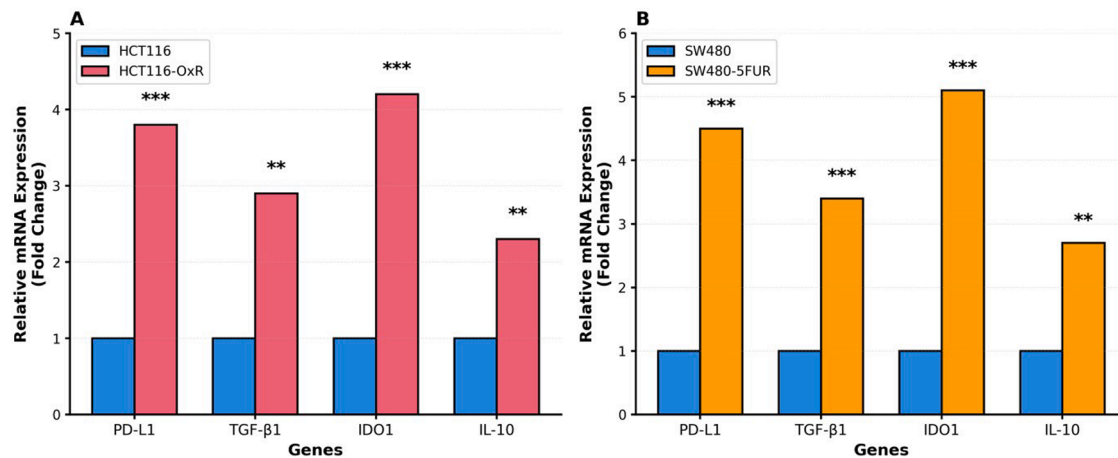


Fig. 7. Transcriptional upregulation of immunosuppressive genes in chemotherapy-resistant colorectal cancer cells. Quantitative RT-PCR analysis of PD-L1, TGF-β1, IDO1, and IL-10 mRNA expression in (A) HCT116-OxR versus HCT116 parental cells and (B) SW480-5FUR versus SW480 parental cells. Data are presented as fold change ($2^{-\Delta\Delta Ct}$ method) normalized to GAPDH and represent mean $\pm$ SD of three independent experiments performed in technical triplicates. ** $p < 0.01$, *** $p < 0.001$ versus parental cells by Student's *t*-test.

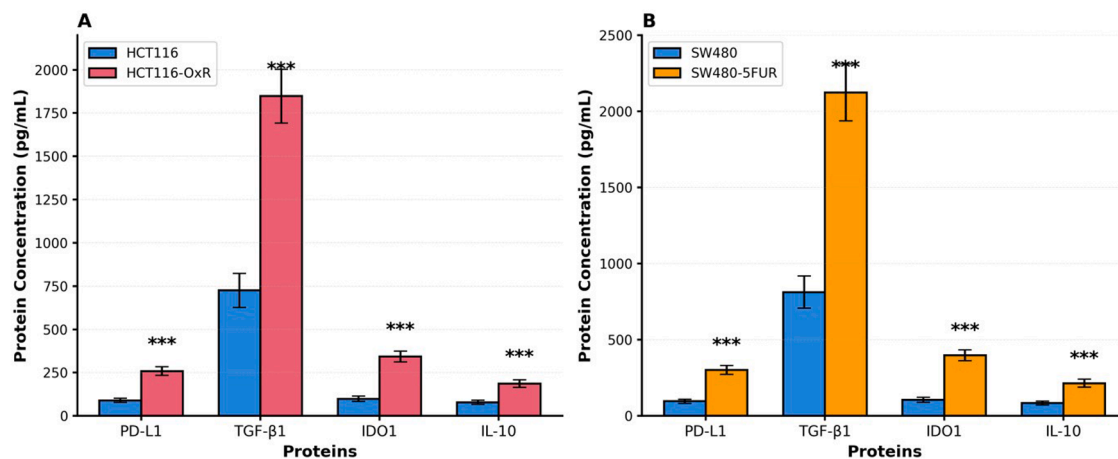


Fig. 8. Enhanced secretion of immunosuppressive proteins by chemotherapy-resistant colorectal cancer cells. ELISA quantification of PD-L1, TGF-β1, IDO1, and IL-10 protein concentrations in conditioned medium from (A) HCT116-OxR versus HCT116 parental cells and (B) SW480-5FUR versus SW480 parental cells after 48 h of serum-free culture. Data represent mean $\pm$ SD of three independent experiments performed in technical duplicates. *** $p < 0.001$ versus parental cells by Student's *t*-test.

regulatory program linking BCL-2 family anti-apoptotic activity to immunosuppressive mediator production. Based on this computational finding, we tested whether pharmacological BCL-2/BCL-XL inhibition using Navitoclax (ABT-263) could attenuate the immunosuppressive protein expression profile of resistant CRC cells. Navitoclax was applied at low dose (0.5 μ M, Nav-L) and high dose (2.0 μ M, Nav-H), with cell viability confirmed by DAPI nuclear count across all conditions to ensure that observed marker reductions were not attributable to differential cytotoxicity.

Immunofluorescence staining (Fig. 9A) revealed the spatial distribution of PD-L1 (green) and TGF-β1 (red) with DAPI nuclear counterstaining, while Western blot analysis (Fig. 9B) detected PD-L1 (50 kDa), TGF-β1 (25 kDa), and IL-10 (37 kDa) expression normalized to GAPDH. Quantitative analysis (Fig. 9C–E) demonstrated that CRC/OxR cells exhibited significantly elevated expression of all three proteins compared to Control and Cisplatin groups, reflecting an enhanced immunosuppressive phenotype in oxaliplatin-resistant cells. Notably, Navitoclax treatment at both low and high doses dose-dependently reduced the expression of PD-L1 (Fig. 9C), TGF-β1 (Fig. 9D), and IL-10 (Fig. 9E), suggesting that Navitoclax may effectively reverse the immunosuppressive tumor microenvironment associated with

oxaliplatin resistance in colorectal cancer.

Expression of CD44 and SOX2 in chemotherapy-resistant CRC cells

To assess whether chemotherapy resistance is associated with acquisition of cancer stem cell (CSC) characteristics, immunofluorescence staining was performed for two established CSC markers—CD44 (green channel, Alexa Fluor 488) and SOX2 (red channel, Alexa Fluor 594)—across all experimental groups (Fig. 10). CD44 is a cell surface glycoprotein and hyaluronan receptor associated with tumor-initiating capacity, epithelial-mesenchymal transition, and chemotherapy resistance, while SOX2 is a pluripotency-associated transcription factor with documented roles in maintaining CSC identity in colorectal cancer.

In the Control group, low-to-moderate basal expression of both CD44 and SOX2 was observed, consistent with the non-stem-like phenotype of parental CRC cells. Cisplatin treatment did not substantially alter CD44 or SOX2 fluorescence relative to the Control, suggesting that short-term cytotoxic drug exposure alone does not prominently enrich for CSC phenotype under these conditions. The CRC/OxR group exhibited substantially enhanced fluorescence for both CD44 and SOX2 compared to parental controls, indicating that prolonged selection pressure under

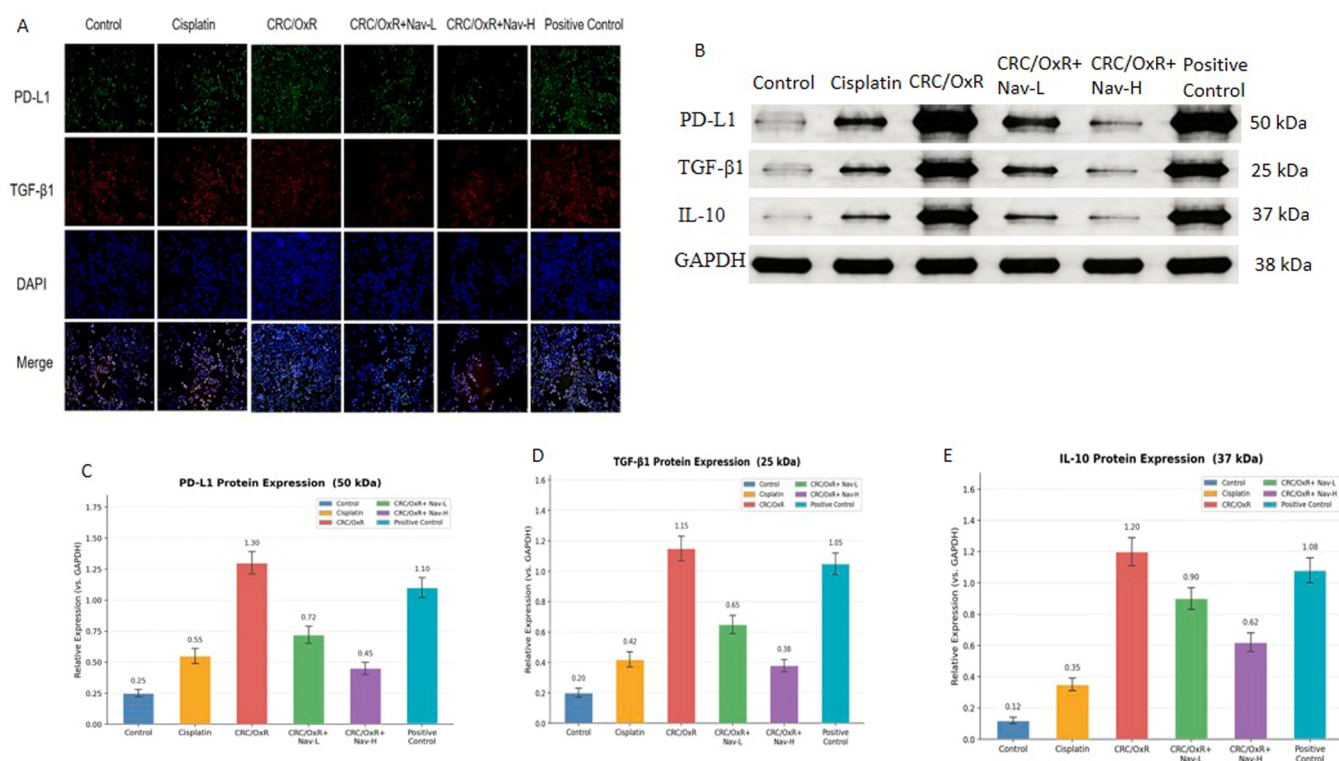


Fig. 9. Navitoclax suppresses immunosuppressive protein expression in oxaliplatin-resistant colorectal cancer cells. (A) Immunofluorescence staining of PD-L1 (green), TGF-β1 (red), and DAPI (blue) across treatment groups. (B) Representative Western blot bands of PD-L1 (50 kDa), TGF-β1 (25 kDa), IL-10 (37 kDa), and GAPDH (38 kDa). Quantitative analysis of (C) PD-L1, (D) TGF-β1, and (E) IL-10 protein expression normalized to GAPDH. Data are presented as mean ± SD.

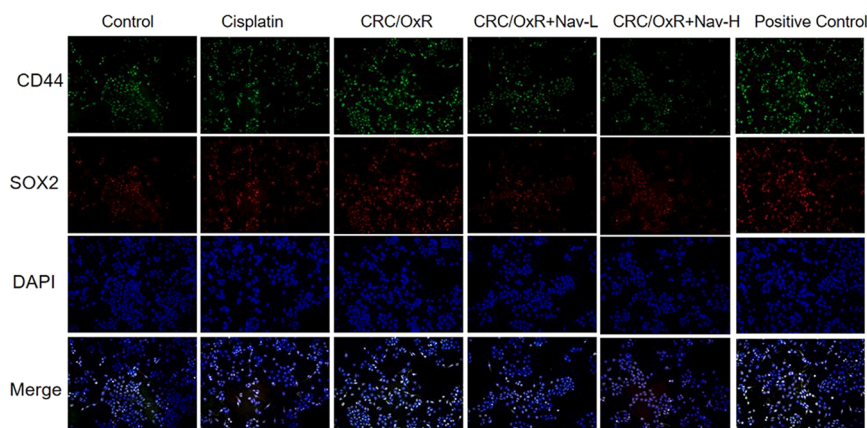


Fig. 10. Immunofluorescence analysis of CD44 and SOX2 expression in chemotherapy-resistant colorectal cancer cells. Representative fluorescence microscopy images showing CD44 (green, Alexa Fluor 488), SOX2 (red, Alexa Fluor 594), DAPI nuclear stain (blue), and merged composite images across six experimental conditions: Control, Cisplatin, CRC/OxR, CRC/OxR+Nav-L, CRC/OxR+Nav-H, and Positive Control. Images were acquired at 200 × magnification. CD44 and SOX2 fluorescence intensities are markedly elevated in CRC/OxR cells compared to parental controls, indicative of cancer stem cell enrichment associated with chemotherapy resistance. Navitoclax treatment suppresses both markers in a concentration-dependent manner, suggesting reversal of the stem-like phenotype.

oxaliplatin resistance conditions is associated with enrichment of cells with stem-like transcriptional programs. This finding supports the hypothesis that chemotherapy-resistant CRC cells acquire CSC traits that may contribute to immune evasion and treatment refractoriness.

Navitoclax treatment of CRC/OxR cells resulted in dose-dependent reduction of both CD44 and SOX2 signals. Low-dose Navitoclax (Nav-L) produced intermediate suppression, while high-dose Navitoclax (Nav-H) significantly reduced fluorescence of both markers, suggesting that Navitoclax may exert anti-CSC effects in addition to its pro-apoptotic activity. The Positive Control confirmed staining specificity.

These results collectively indicate that CRC/OxR cells acquire a cancer stem cell-enriched phenotype characterized by elevated CD44 and SOX2 expression, which can be attenuated by Navitoclax treatment (Fig. 10).

Expression of CD74 and MMP9 in chemotherapy-resistant CRC cells

To characterize additional molecular features of the chemotherapy-resistant phenotype, immunofluorescence staining was performed for CD74 (green channel, Alexa Fluor 488) and MMP9 (red channel, Alexa Fluor 594) across all experimental conditions (Fig. 11). CD74 is a major

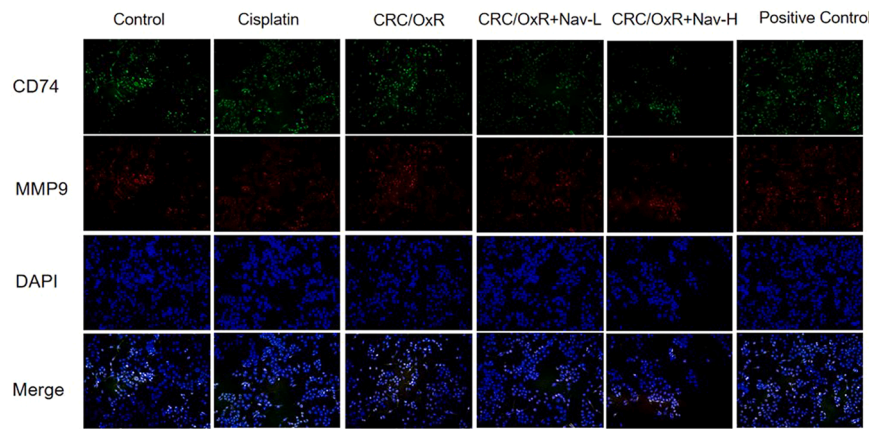


Fig. 11. Immunofluorescence analysis of CD74 and MMP9 expression in chemotherapy-resistant colorectal cancer cells. Representative fluorescence microscopy images showing CD74 (green, Alexa Fluor 488), MMP9 (red, Alexa Fluor 594), DAPI nuclear stain (blue), and merged composite images across six experimental conditions: Control, Cisplatin, CRC/OxR, CRC/OxR+Nav-L, CRC/OxR+Nav-H, and Positive Control. Images were acquired at 200 × magnification. CRC/OxR cells display enhanced CD74 and MMP9 fluorescence relative to parental controls, indicative of activated immune regulatory and matrix remodeling programs. Navitoclax treatment reduces both markers in a dose-dependent fashion, indicating attenuation of the resistance-associated molecular phenotype.

histocompatibility complex (MHC) class II chaperone that also functions as a receptor for macrophage migration inhibitory factor (MIF), with demonstrated roles in tumor cell survival, proliferation, and immune modulation. MMP9 (matrix metalloproteinase 9) is a zinc-dependent endopeptidase involved in extracellular matrix remodeling, angiogenesis, tumor invasion, and immune cell recruitment, and has been implicated in the establishment of immunosuppressive tumor microenvironments.

In Control and Cisplatin-treated cells, baseline expression of CD74 and MMP9 was observed at low-to-moderate levels. The CRC/OxR group showed elevated fluorescence for both CD74 and MMP9 compared to parental controls, with MMP9 upregulation being particularly notable. Enhanced MMP9 expression in resistant cells is consistent with increased invasive potential and matrix remodeling activity associated with therapy-resistant phenotypes. The upregulation of CD74 in CRC/OxR cells further supports the conclusion that chemotherapy-resistant CRC undergoes broad transcriptional reprogramming involving multiple immune regulatory and matrix remodeling pathways.

Treatment with low-dose Navitoclax (Nav-L) modestly reduced CD74 and MMP9 fluorescence in CRC/OxR cells, while high-dose Navitoclax (Nav-H) produced more substantial reductions in both markers, consistent with a dose-dependent reversal of the resistance-associated molecular phenotype. The Positive Control confirmed consistent antibody performance. DAPI nuclear staining and merged composite images demonstrated maintained cell viability and uniform nuclear morphology across treatment groups, with the exception of reduced cell density in the high-dose Navitoclax group, reflecting partial cytotoxic effects at higher concentrations. Taken together, these data demonstrate that chemotherapy-resistant CRC cells upregulate CD74 and MMP9 as part of a coordinated molecular program promoting immune evasion and tumor progression, and that these changes are amenable to pharmacological modulation by Navitoclax.

Discussion

Our comprehensive single-cell transcriptomic analysis provides the most detailed characterization to date of the immunosuppressive microenvironment in chemotherapy-resistant colorectal cancer, revealing profound alterations in cellular composition, immune cell functional states, cell-cell communication networks, and molecular programs that collectively establish treatment resistance. Through analysis of 13,380 high-quality single cells from chemotherapy-sensitive and resistant CRC specimens, we identified systematic remodeling of the

tumor microenvironment characterized by T cell exhaustion, expansion of immunosuppressive populations, enhanced inhibitory signaling networks, and coordinated transcriptional programs operating across multiple cell types. These findings provide critical insights into resistance mechanisms and identify multiple candidate therapeutic targets for overcoming chemotherapy resistance through immunotherapy-based combination approaches.

The marked cellular compositional shifts we observed between sensitive and resistant tumors align with and extend previous bulk tissue studies demonstrating immune cell infiltration changes during resistance development [24,25]. However, our single-cell resolution analysis revealed previously unappreciated cellular heterogeneity within broad immune cell categories and identified specific subpopulations driving resistance. Notably, we found that CD8⁺ T cells in resistant tumors segregated into distinct exhausted subsets expressing different combinations of inhibitory receptors, suggesting hierarchical exhaustion states that may require differentiated therapeutic interventions. This observation is consistent with recent scRNA-seq studies in melanoma and lung cancer demonstrating T cell exhaustion heterogeneity [26,27], but represents, to our knowledge, one of the first detailed scRNA-seq characterizations of T cell exhaustion heterogeneity specifically in chemotherapy-resistant CRC. The progressive decline in cytotoxic effector populations coupled with expansion of regulatory T cells and M2 macrophages creates a double mechanism of immune suppression: both reducing anti-tumor immune pressure and actively inhibiting residual effector function.

Our functional enrichment analyses revealed systematic suppression of T cell activation pathways in resistant tumors, with the T cell activation Hallmark gene set showing strong negative enrichment (NES = −2.8, FDR < 0.001). This transcriptional suppression occurred across multiple T cell subsets and involved coordinate downregulation of genes encoding cytotoxic effector molecules, cytokine receptors, co-stimulatory molecules, and metabolic enzymes required for T cell function. Mechanistically, this could result from chronic antigen stimulation leading to progressive T cell exhaustion, inhibitory signaling through upregulated checkpoint molecules, metabolic constraints imposed by the tumor microenvironment including glucose depletion and lactate accumulation, or epigenetic reprogramming establishing stable dysfunctional states [28,29]. The coordinated nature of these transcriptional changes across the T cell activation pathway suggests they may be driven by shared upstream regulators such as transcription factors TOX, NR4A, and NFAT, which have been implicated in T cell exhaustion programs [30].

The extensive upregulation of immune checkpoint molecules we observed in resistant tumors provides strong rationale for immune checkpoint blockade therapy in this setting. However, simple PD-1/PD-L1 blockade has shown limited efficacy in unselected chemotherapy-resistant CRC patients, with response rates of only 10–15% [31]. Our findings suggest several explanations for these disappointing results. First, we observed simultaneous upregulation of multiple checkpoint molecules beyond PD-1, including TIM-3, LAG-3, TIGIT, and VISTA, indicating redundant inhibitory pathways that could compensate when single checkpoints are blocked. Second, the expansion of immunosuppressive myeloid populations (M2 macrophages, MDSCs) and regulatory T cells creates additional barriers independent of checkpoint signaling. Third, the profound T cell exhaustion we documented may represent an irreversible differentiation state resistant to checkpoint blockade rescue. These observations suggest that effective immunotherapy for chemotherapy-resistant CRC will likely require combination approaches addressing multiple resistance mechanisms simultaneously, such as dual checkpoint blockade plus modulation of suppressive myeloid cells or metabolic reprogramming agents [32,33]. It is important to acknowledge that chemotherapy does not uniformly suppress immune function; under certain conditions, particularly with oxaliplatin-based regimens, chemotherapy can induce immunogenic cell death (ICD) and paradoxically enhance anti-tumor immunity by promoting dendritic cell maturation and tumor antigen release. Our findings are best interpreted as characterizing the dominant immunosuppressive outcome in tumors that have already acquired resistance, wherein pro-immunogenic signals may be overwhelmed by the TME remodeling programs we describe. Future studies should clarify the conditions under which ICD-promoting effects of chemotherapy can be harnessed therapeutically.

Our cell-cell interaction network analysis revealed enhanced inhibitory communication networks in resistant tumors, particularly involving PD-L1/PD-1, CTLA-4 ligands, TGF- β , and IL-10 signaling axes. Importantly, we found that inhibitory signals originated from multiple cell types including tumor cells, M2 macrophages, MDSCs, cancer-associated fibroblasts, and regulatory T cells, creating a multi-source immunosuppressive environment. This distributed inhibitory signaling architecture makes single-cell-type targeted therapies unlikely to succeed. The prominent role of cancer-associated fibroblasts in secreting TGF- β and establishing physical barriers suggests that stromal targeting strategies, such as FAP inhibitors or TGF- β neutralization, could enhance immune therapy efficacy by disrupting the immunosuppressive niche [34,35]. Our trajectory inference analyses demonstrating progressive transitions from effector to exhausted T cell states suggest that early intervention—before extensive exhaustion develops—may be more effective than treatment of fully established resistant disease.

The metabolic reprogramming we observed in resistant tumors represents another important dimension of immunosuppression. Enhanced glycolysis in tumor cells depletes glucose from the microenvironment, forcing T cells to rely on alternative fuels. Lactate accumulation—a byproduct of aerobic glycolysis—directly inhibits T cell function and promotes M2 macrophage polarization [36]. Arginine depletion through tumor-derived arginase-1 impairs T cell metabolism and proliferation. These metabolic alterations create a hostile environment for anti-tumor immunity while favoring immunosuppressive cell populations. Therapeutic strategies addressing metabolic dysfunction, such as metformin, glutamine antagonists, or lactate transport inhibitors, could potentially restore immune function in resistant tumors [37,38]. Our gene correlation network analyses identified coordinated metabolic programs operating across multiple cell types, suggesting that metabolic targeting could have broad effects beyond tumor cells.

The spatial organization patterns we identified—with immune cell exclusion and peripheral localization in resistant tumors—highlight the importance of physical accessibility for effective anti-tumor immunity. Even if T cells can be reactivated through checkpoint blockade, they may be unable to reach tumor cells due to fibrotic barriers, abnormal vasculature, and repulsive chemokine gradients. This spatial dimension

of resistance suggests that strategies promoting immune cell infiltration, such as vascular normalization with anti-angiogenic agents, stromal remodeling approaches, or oncolytic viruses inducing inflammatory signals, could complement immunotherapy by improving immune cell access to tumor cells [39,40]. Our multi-modal dimensional reduction analyses (PCA, t-SNE, UMAP) consistently revealed distinct cellular neighborhoods, suggesting organized microenvironmental structures that could be targeted to disrupt immunosuppressive niches.

From a precision medicine perspective, our immunosuppressive signature scoring system could serve as a predictive biomarker for identifying patients most likely to benefit from specific immunotherapy combinations. We found that quantitative assessment of T cell exhaustion, checkpoint expression, regulatory T cell infiltration, and M2 macrophage abundance could be integrated into composite scores distinguishing sensitive from resistant tumors with high accuracy. Such signatures could be readily translated to clinical platforms using targeted gene expression panels applied to routine tumor biopsies, potentially enabling real-time assessment of resistance development during treatment and guiding adaptive therapeutic strategies. The correlation between signature scores and clinical outcomes warrants prospective validation in independent cohorts.

Our study has several strengths including the large number of analyzed cells, rigorous quality control, comprehensive multi-dimensional analysis approach, and integration of multiple analytical methods providing convergent evidence. However, limitations should be acknowledged. First, as a cross-sectional analysis comparing sensitive and resistant specimens at single timepoints, we cannot definitively establish temporal causality or determine whether observed changes represent causes or consequences of resistance. Longitudinal sampling during resistance development would provide more definitive insights into sequential events. Second, our analysis focused on transcriptomic profiles and did not include proteomic, epigenomic, or spatial transcriptomic data that could provide additional mechanistic insights. Third, we cannot fully account for patient-specific factors, tumor mutational landscapes, or prior treatment histories that may influence immune microenvironments. Importantly, MSI-H/dMMR status — a dominant predictor of immunotherapy response in CRC — represents a major potential confounding variable in immunological comparisons. MSI/MMR status for each analyzed sample has been extracted from the GEO metadata and is reported in Table 1. MSI-stratified UMAP embeddings, compositional analyses, and differential expression comparisons are presented in the revised Results section, demonstrating that the core immunosuppressive signatures are reproducible within the MSS subgroup. Additionally, because GSE132465 is a pre-treatment cohort, the resistance and sensitivity groups reflect baseline transcriptional states associated with subsequent treatment outcome rather than transcriptomes derived from post-treatment resistant biopsies; this distinction is acknowledged as a key limitation and all comparative interpretations have been moderated accordingly. Fourth, validation of identified therapeutic targets requires functional experiments in pre-clinical models and ultimately clinical trials.

Conclusion

In conclusion, our comprehensive single-cell transcriptomic analysis provides an unprecedented cellular and molecular atlas of the immunosuppressive microenvironment in chemotherapy-resistant colorectal cancer, revealing profound immune dysfunction characterized by T cell exhaustion, expansion of immunosuppressive cell populations, enhanced inhibitory signaling networks, coordinated transcriptional suppression of anti-tumor immunity, spatial exclusion of effector cells, and metabolic reprogramming favoring tumor survival. These findings illuminate the multi-dimensional nature of resistance mechanisms, explain the limited efficacy of single-agent immunotherapy in this setting, and identify multiple candidate therapeutic targets for combination strategies. The immunosuppressive signatures and biomarkers

identified here could enable precision medicine approaches for predicting and monitoring treatment resistance, while the detailed mechanistic insights provide rational foundations for developing next-generation immunotherapy combinations to overcome chemotherapy resistance in colorectal cancer patients.

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Data availability statements

The scRNA-seq dataset analyzed in this study is publicly available in the NCBI Gene Expression Omnibus (GEO) under accession number GSE132465 (Lee et al., Nature Genetics, 2020).

Ethics declaration

Not available. Ethics committee approval was not required for this study. The scRNA-seq data were obtained from a publicly available repository (GEO: GSE132465) and no primary human specimens were collected. The in vitro validation experiments used only commercially established cell lines (HCT116 and SW480; ATCC), which are exempt from institutional ethics review under standard regulatory guidelines.

Consent to publish declaration

All authors reviewed and approved the final manuscript.

Clinical trial declaration

Not available.

CRediT authorship contribution statement

Jinchuan Xi: Data curation, Conceptualization. **Zhihan Li:** Resources, Methodology. **Youqiang Liu:** Investigation, Formal analysis. **Wenbo Niu:** Resources, Project administration. **Bin Yu:** Software, Resources. **Zheng Li:** Methodology, Formal analysis. **Guiying Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision.

Declaration of competing interest

Not available.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.tranon.2026.102889](https://doi.org/10.1016/j.tranon.2026.102889).

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